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A Stochastic Wavelet Emulator of 2D Navier–Stokes

Biorthogonal wavelet–Galerkin discretisation and guided score-based modelling of the transition kernel

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Outline

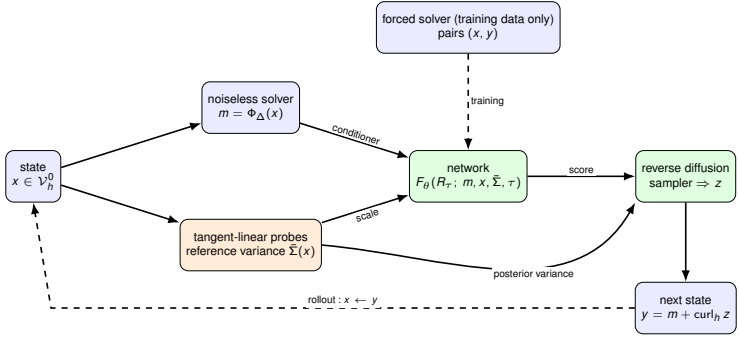
1. Problem and method
2. Wavelet–Galerkin discretisation
3. Learning problem and reference
4. Score-based model
5. Sampling
6. Results
7. Conclusion

Objective : learn a stochastic transition kernel

Stochastically forced 2D incompressible Navier–Stokes on $\mathbb{T}^2$, discretised in space and time : a Markov chain on $\mathcal{V}_h^0$, the space of discrete divergence-free, zero-mean velocity fields. **Goal.** For a lag Δ, learn to sample

$$y \sim p_\Delta(\cdot \mid x), \qquad x = \text{state at } t, \quad y = \text{state at } t + \Delta,$$

and chain the samples into long trajectories that reproduce the statistics of the solver.



Stochastically forced Navier–Stokes on the torus

On $\mathbb{T}^2 = [0,1)^2$, $\boldsymbol{\xi} = (\xi_1, \xi_2)$, velocity $\boldsymbol{u}(\boldsymbol{\xi}, t)$, pressure p :

$$d\boldsymbol{u} + \big[(\boldsymbol{u}\cdot\nabla)\boldsymbol{u} + \nabla \mathrm{p} - \nu\Delta\boldsymbol{u} + \mu_d \boldsymbol{u}\big]dt = d\boldsymbol{W}_t, \qquad \nabla\cdot\boldsymbol{u} = 0,$$

$\boldsymbol{W}_t$: divergence-free, zero-mean $\mathcal{Q}$-Wiener process, $\mathrm{Tr}\,\mathcal{Q} < \infty$, on $H := \{\boldsymbol{u} \in L^2(\mathbb{T}^2)^2 : \nabla\cdot\boldsymbol{u} = 0,\ \int \boldsymbol{u} = 0\}$.

Energy balance. (Itô ; $E = \frac{1}{2}\|\boldsymbol{u}\|_{L^2}^2$, $Z = \frac{1}{2}\|\nabla\boldsymbol{u}\|_{L^2}^2$ and because $\int(\boldsymbol{u}\cdot\nabla\boldsymbol{u})\cdot\boldsymbol{u} = 0$ and $\int \nabla\mathrm{p}\cdot\boldsymbol{u} = 0$.)

$$dE = \big(-2\mu_d E - 2\nu Z + \varepsilon_{\mathrm{inj}}\big)dt + \langle \boldsymbol{u}, d\boldsymbol{W}_t\rangle, \qquad ,$$

In a stationary state

$$\boxed{2\mu_d\langle E\rangle + 2\nu\langle Z\rangle = \varepsilon_{\mathrm{inj}}}$$

forcing	without it $p_\Delta(\cdot \mid x)$ is a Dirac mass (no density, no score) and the flow decays
Ekman drag $\mu_d > 0$	fixes the large-scale energy, $\langle E\rangle \le \varepsilon_{\mathrm{inj}}/(2\mu_d)$; arrests the inverse cascade at $(\varepsilon_{\mathrm{inj}}/\mu_d^3)^{1/2}$
zero-mean forcing	$\frac{d}{dt}\int \boldsymbol{u} = -\mu_d \int \boldsymbol{u}$: no mean flow, hence the state space H

Temam (2001) ; Da Prato–Zabczyk (1992) ; Flandoli (1994) ; Kuksin–Shirikyan (2012) ; Boffetta–Ecke (2012).

Biorthogonal multiresolution and three coordinate systems

CDF 9/7 (`bior4.4`) : primal and dual scaling functions $\varphi, \tilde{\varphi}$, $\langle \tilde{\varphi}, \varphi(\cdot - l) \rangle = \delta_{l0}$. Level $J : n = 2^J$, $h = 2^{-J}$, $\varphi_{J,k} = \varphi(2^J \cdot - k)$ periodised,

$$V_J = V_{j_{\min}} \oplus W_{j_{\min}} \oplus \cdots \oplus W_{J-1}.$$

Analysis W_{fwd} (dual filters), synthesis W_{inv} (primal filters) : $W_{\mathrm{fwd}} W_{\mathrm{inv}} = I$, but W_{fwd} is *not* orthogonal.



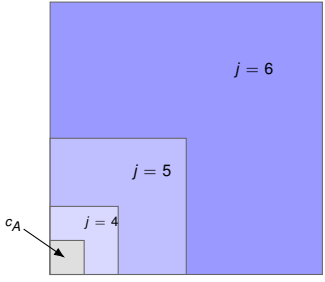
scaling $\mathrm{u} = (U_s, V_s) \in \mathbb{R}^{2n^2}$
physics : banded circulants

$\mathbb{W} = I_2 \otimes W_{\mathrm{fwd}} \otimes W_{\mathrm{fwd}}$

wavelet $x = (\mathrm{vec}\, U_w, \mathrm{vec}\, V_w) \in \mathbb{R}^{2n^2}$
statistics : levels, orbits

L^2 metric in any coordinates : $\|\boldsymbol{u}_h\|_{L^2}^2 = h^2\, \mathrm{u}^\top G_2 \mathrm{u} = h^2\, x^\top G_{2,w} x$, $G_{2,w} = \mathbb{W}^{-\top} G_2 \mathbb{W}^{-1}$.

Levels, orbits and the dyadic symmetry group



$2^J \times 2^J$ coefficients, $J = 7$

Level map (max rule). $j(i_1, i_2) = \max(l(i_1), l(i_2))$, c_A labelled $j_{\min} - 1$: fine along one axis means fine. $|\mathcal{I}_j| = 3 \cdot 4^j$ ($j \geq j_{\min}$), $|\mathcal{I}_{j_{\min}-1}| = 4^{j_{\min}}$.

Orbits. level-j positions equal modulo $2^{j-j_{\min}}$: $n_{\mathrm{orb}} = 4^{J-j_{\min}}$ orbits (64 at $J = 7$) of size $4^{j_{\min}}$ per scalar field.

connection coefficients $\Rightarrow$ circulant operators

Bubnov–Galerkin (test space = trial space = V_J). Two-term coefficients

$$m_l = \langle \varphi, \varphi(\cdot - l) \rangle, \qquad d_l = \langle \varphi, \varphi'(\cdot - l) \rangle, \qquad k_l = \langle \varphi, \varphi''(\cdot - l) \rangle.$$

Proposition. $\gamma^{(q)} \in \{m, d, k\}$ $(q = 0, 1, 2)$ is the eigenvector $\mathcal{T}\gamma^{(q)} = 2^{-q}\gamma^{(q)}$ of a matrix built from the refinement filter, normalised by polynomial reproduction ($\sum_l m_l = 1$, $\sum_l l\, d_l = 1$, $\sum_l l^2 k_l = 2$). Triad coefficients : same, eigenvalues 2 and 1. No quadratures.

Operator	Definition	Meaning	Structure
M	$\mathrm{circ}(m) = G$	mass = Gram (Bubnov)	$M = M^\top \succ 0$
D	$2^J \mathrm{circ}(d)$	derivative	$D^\top = -D$
K	$2^{2J} \mathrm{circ}(k)$	weak second derivative	$K = K^\top \preceq 0$
$M_2,\ C$	$M \otimes M,\ [\, D \otimes M \ \ M \otimes D \,]$	2D mass, weak divergence	$\mathrm{ran}\, C^\top$ = weak gradients
$\mathcal{K}$	$K \otimes M + M \otimes K$	weak Laplacian	$M_2^{-1}\mathcal{K} = M^{-1}K \oplus M^{-1}K$

All 1D operators are commuting circulants, diagonalised by the FFT ; 2D operators lie in the commutative algebra of circulant $\otimes$ circulant. Nothing $n^2 \times n^2$ is formed.

energy-neutral advection

Skew-symmetric (Temam) trilinear form :

$$b(\boldsymbol{u}, \boldsymbol{w}, \boldsymbol{\psi}) = \int_{\mathbb{T}^2} (\boldsymbol{u} \cdot \nabla)\boldsymbol{w} \cdot \boldsymbol{\psi} + \tfrac{1}{2} \int_{\mathbb{T}^2} (\nabla \cdot \boldsymbol{u})\, \boldsymbol{w} \cdot \boldsymbol{\psi}.$$

Discrete weak advection $\mathcal{N}(\mathrm{u}; \mathrm{w})$ (one entry per test function $\varphi_p \otimes \varphi_q$), evaluated with the triad coefficients $\langle \varphi, \varphi(\cdot - p)\varphi(\cdot - q)\rangle$, $\langle \varphi, \varphi(\cdot - p)\varphi'(\cdot - q)\rangle$.

Proposition (Bubnov). For all $\boldsymbol{u}_h, \boldsymbol{w}_h \in V_J^2$: $b(\boldsymbol{u}_h, \boldsymbol{w}_h, \boldsymbol{w}_h) = 0$, hence

$$\mathrm{w}^\top \mathcal{N}(\mathrm{u}; \mathrm{w}) = 0 \quad \text{for every } \mathrm{u}, \text{ divergence free or not.}$$

Idea : $\int (\boldsymbol{u} \cdot \nabla \boldsymbol{w}) \cdot \boldsymbol{w} = -\frac{1}{2} \int (\nabla \cdot \boldsymbol{u}) |\boldsymbol{w}|^2$, and $\boldsymbol{w}_h$ is itself an admissible test function.

Consequences.

- advection drops out of the discrete energy identity and of the Lyapunov bound ;
- the mean flow is not changed by advection.

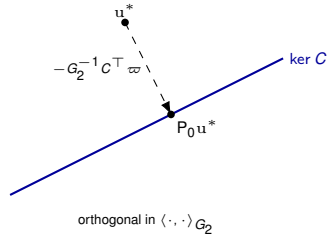
Incompressibility : the discrete Leray projector

Saddle point (scaling basis, pressure ϖ) :

$$M_2\dot{\mathrm{u}} = \nu\mathcal{K}\mathrm{u} - \mu_d M_2\mathrm{u} - \mathcal{N}(\mathrm{u}) - C^\top\varpi + \text{forcing}, \qquad C\mathrm{u} = 0.$$

Eliminating ϖ is projecting :

$$\mathrm{P}_0 = I - G_2^{-1}C^\top\mathcal{S}^+C, \qquad \mathcal{S} = CG_2^{-1}C^\top.$$



Proposition. P_0 is the G_2-orthogonal projector onto $\ker C$:
$\mathrm{P}_0\mathrm{u}^* = \arg\min_{C\mathrm{v}=0}\|\mathrm{v}-\mathrm{u}^*\|_{G_2}$, $G_2\mathrm{P}_0 = \mathrm{P}_0^\top G_2$. $\mathcal{S}^+$ is from two $n\times n$ eigenproblems.

Projected dynamics :

$$\dot{\mathrm{u}} = L\mathrm{u} + \mathcal{H}(\mathrm{u}) + \text{forcing}, \qquad L = \nu(M^{-1}K \oplus M^{-1}K) - \mu_d I, \qquad \mathcal{H}(\mathrm{u}) = -\mathrm{P}M_2^{-1}\mathcal{N}(\mathrm{u})$$

Lemma (no splitting error). P lies in the commutative algebra : $\mathrm{P}L = L\mathrm{P}$, $\mathrm{P}e^{tL} = e^{tL}\mathrm{P}$. Projecting once per stage is exact, not a fractional step (measured commutator 3×10^{-13}).

Stochastic forcing, drawn in stream-function coordinates

Discrete stream function : unconstrained coordinates of $\mathcal{V}_h^0$ Scaling coordinates, $\Psi \in \mathbb{R}^{n\times n}$:

$$\boxed{\operatorname{curl}_h \Psi := \mathcal{F}\big(\Psi(M^{-1}D)^\top,\ -(M^{-1}D)\Psi\big)} \qquad \text{(discrete } \nabla^\perp\psi\text{)}$$

The noise is drawn as one scalar field, $\zeta \sim \mathcal{N}(0, I_{n^2})$, and mapped by the same curl that reconstructs y :

$$\boxed{B_F := \operatorname{curl}_h\, Q_\psi^{1/2}, \qquad Q_\psi = \operatorname{diag}\big(4^j q_{j(i)}\big)_{i\le n^2}, \qquad q_j = q_0\, 2^{-j(3+\alpha_F)}, \qquad Q := B_F B_F^\top}$$

Proposition. $\operatorname{ran} B_F = \mathcal{V}_h^0$: the noise is divergence free, zero mean and Nyquist free *without any projection*. Hence $Q \succ 0$ on $\mathcal{V}_h^0$ (needed for a density), $PQ = Q$, and Q is $\mathcal{G}$-invariant.

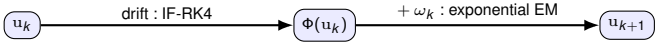
Spectrum. A level-j L^2-normalised stream wavelet has $|\nabla\psi_\lambda| \sim 2^j$: the velocity variance per coefficient is $4^j q_j = q_0\, 2^{-j(1+\alpha_F)}$, so the trace is bounded uniformly in J iff $\sum_j 2^{j(1-\alpha_F)} < \infty \iff \alpha_F > 1$. Production : $\alpha_F = 5$ (large-scale forcing).

Injection rate, exact. $\varepsilon_{\mathrm{inj}} = \frac{1}{2}h^2 \operatorname{Tr}(G_{2,w} Q)$, computed with one noise column per orbit (no Monte Carlo) ; it is linear in q_0, so q_0 is set to prescribe $\varepsilon_{\mathrm{inj}}$.

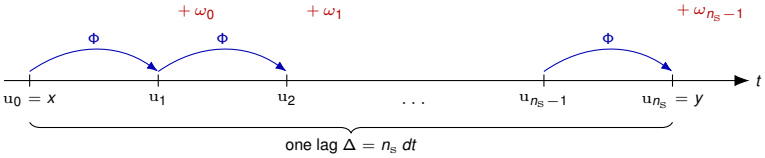
`make_forcing_spec(coords='stream'), make_noise_fn, estimate_eps_inj`

The lag, the residual, and why its law is singular

Reduced SDE on $\mathcal{V}_h^0$ (wavelet coordinates ; L, $\mathcal{H}$ conjugated by $\mathbb{W}$) : $dx = \big(Lx + \mathcal{H}(x)\big)\,dt + B_F\,d\mathcal{B}_t$.



$$u_{k+1} = \Phi(u_k) + \omega_k, \qquad \omega_k = \sqrt{dt}\,\mathrm{f}(dt\,L)\,B_F\,\zeta_k \sim \mathcal{N}(0,\, dt\,\widetilde{Q}), \qquad \mathrm{f}(z) = \sqrt{\frac{e^{2z}-1}{2z}}, \quad \widetilde{Q} := \mathrm{f}\,Q\,\mathrm{f}^\top.$$



$$m(x) := \Phi_\Delta(x) = \Phi^{n_s}(x), \qquad r := \mathrm{curl}_h^+\big(y - m(x)\big) \in \mathcal{R}, \qquad y = m(x) + \mathrm{curl}_h\, r.$$

- $\mathrm{curl}_h^+ : \mathcal{V}_h^0 \to \mathcal{R}$ is a linear bijection : learning the law of r is learning $p_\Delta(\cdot \mid x)$;
- m is $O(1)$; $r = O(\sqrt{\varepsilon_{\mathrm{inj}}\Delta})$, and $r \to 0$ as the forcing vanishes ;
- every $y = m + \mathrm{curl}_h\, r$ is divergence free and zero mean : no projection in the generative model.

Lisa's comment : PB $\dim \mathcal{R} = n^2 - 2n < d_\psi = n^2$: the law of r is singular (codimension $n_{\ker} = 2n$), whereas a diffusion model needs a density on $\mathbb{R}^{d_\psi}$?

The conditional law is Gaussian to leading order

Scale the forcing, $Q_\psi \to \eta_Q Q_\psi$, i.e. $\boldsymbol{\omega} = \eta_Q^{1/2}\hat{\boldsymbol{\omega}}$ with $\hat{\boldsymbol{\omega}}$ fixed. Since $\mathscr{R}_x$ is a polynomial,

$$r = \eta_Q^{1/2}\, \mathrm{J}_x\hat{\boldsymbol{\omega}} + \eta_Q\, \mathrm{B}_x(\hat{\boldsymbol{\omega}}) + O(\eta_Q^{3/2}).$$

Theorem. For fixed x, as $\eta_Q \to 0$:

1. $\eta_Q^{1/2}\mathrm{J}_x\hat{\boldsymbol{\omega}} \sim \mathcal{N}(0, \Sigma(x))$, with $\Sigma(x) \succ 0$ on $\mathcal{R}$ and $\Sigma = O(\eta_Q)$;
2. $\mathbb{E}[r \mid x] = \eta_Q\, \mathbb{E}[\mathrm{B}_x(\hat{\boldsymbol{\omega}})] + O(\eta_Q^2)$;
3. $\mathrm{Cov}(r \mid x) = \Sigma(x) + O(\eta_Q^2)$;
4. the third cumulant of $r \mid x$ is $O(\eta_Q^2)$.

Key argument : $\hat{\boldsymbol{\omega}} \overset{d}{=} -\hat{\boldsymbol{\omega}}$ kills every odd power of $\eta_Q^{1/2}$.

In units of the standard deviation. $(O(\eta_Q^{1/2}))$: mean shift and skewness are $O(\eta_Q^{1/2})$, the covariance error is $O(\eta_Q)$ relative.

Corollary. Evaluating the matrix on the noiseless path costs only $O(\eta_Q)$ relative : $\mathbb{E}[\gamma_r(\boldsymbol{\omega})] = \Sigma(x)(1 + O(\eta_Q))$, γ_r the matrix at $\boldsymbol{\omega}$.

Gaussian part : **computed**. Orders ≥ 2 in the noise (noise advecting noise) : **learned**.

One lag as a function of its injected noise

Fix x. For $\boldsymbol{\omega} = (\omega_0, \dots, \omega_{n_s-1}) \in (\mathcal{V}_h^0)^{n_s}$, set $u_0 = x$, $u_{k+1} = \Phi(u_k) + \omega_k$ and

$$\mathscr{Y}_x(\boldsymbol{\omega}) := u_{n_s}, \qquad \mathscr{R}_x(\boldsymbol{\omega}) := \mathrm{curl}_h^+ \big(\mathscr{Y}_x(\boldsymbol{\omega}) - \mathscr{Y}_x(0)\big).$$

Then $m = \mathscr{Y}_x(0)$ (noiseless path x_k°), $y = \mathscr{Y}_x(\boldsymbol{\omega})$, $r = \mathscr{R}_x(\boldsymbol{\omega})$ and $\mathscr{R}_x(0) = 0$.

Discrete derivative. $\mathscr{D}_k \mathscr{R}_x := \partial \mathscr{R}_x / \partial \omega_k = \mathrm{curl}_h^+ \; D\Phi(u_{n_s-1}) \cdots D\Phi(u_{k+1})$; at zero noise $\mathscr{D}_k \mathscr{R}_x(0) = \mathrm{curl}_h^+ \, \mathsf{A}_k$, $\mathsf{A}_k := D\Phi(x^\circ_{n_s-1}) \cdots D\Phi(x^\circ_{k+1})$.

matrix at zero noise.

$$\Sigma(x) := \sum_{k=0}^{n_s-1} \mathscr{D}_k \mathscr{R}_x(0)\, (dt\, \widetilde{Q})\, \mathscr{D}_k \mathscr{R}_x(0)^\top = dt \sum_{k=0}^{n_s-1} \mathrm{curl}_h^+ \, \mathsf{A}_k \, \widetilde{Q} \, \mathsf{A}_k^\top \, \mathrm{curl}_h^{+\top}$$

$\Sigma(x)$ is the covariance of the Taylor-linear part $\mathsf{J}_x \boldsymbol{\omega} = \sum_k \mathrm{curl}_h^+ \, \mathsf{A}_k \omega_k$: how much injected noise reaches each coefficient of r.

Continuum : $\mathscr{D}_{t'} y = Y_{\Delta,t'} B_F$, $\Sigma \to \mathrm{curl}_h^+ \int_0^\Delta Y_{\Delta,t'} Q Y_{\Delta,t'}^\top dt' \; \mathrm{curl}_h^{+\top}$ (Nualart 2006). Not needed : the discrete objects are for the scheme that produces the data.bk-mall

Computing $\Sigma(x)$: offline linear law and tangent-linear probes

Tangent-linear model (TLM). Probe b : draw $\boldsymbol{\omega}^b \sim \mathcal{N}(0, \mathbf{C}_\omega)$ and run

$$\delta^b_0 = 0, \quad \delta^b_{k+1} = D\Phi(x^\circ_k)\,\delta^b_k + \omega^b_k \quad \Longrightarrow \quad \mathrm{curl}^+_h\,\delta^b_{n_\mathrm{s}} = \mathsf{J}_x \boldsymbol{\omega}^b, \qquad \mathbb{E}\big[(\mathrm{curl}^+_h\,\delta^b_{n_\mathrm{s}})^2_i\big] = \boldsymbol{\Sigma}_{ii}(x).$$

Estimator $\widehat{\boldsymbol{\Sigma}}_{ii}(x) = \frac{1}{n_\mathrm{probe}} \sum_b (\mathrm{curl}^+_h\,\delta^b_{n_\mathrm{s}})^2_i$: relative error $\sqrt{2/n_\mathrm{probe}}$ per coefficient ; cost n_s Jacobian–vector products per probe (`jax.jvp`) ; nothing $d_\psi \times d_\psi$ is formed.

Mode	Tangent map	Output
B0 (offline)	$D\Phi \equiv \mathsf{P} e^{dtL}$ (if $\mathcal{H} = 0$)	Σ^{lin} : covariance of r for Stokes+drag ; x-independent, orbit-constant, computed (dyadic block algebra)
TLM (per x)	true $D\Phi(x^\circ_k)$ along the noiseless path	$\widehat{\boldsymbol{\Sigma}}_{ii}(x)$: how the current flow amplifies or damps the injected noise

Fixed probes (common random numbers) : $x \mapsto \widehat{\boldsymbol{\Sigma}}(x)$ is one deterministic smooth function, identical in the dataset, in training and in the emulator.

The diagonal surrogate $\bar{\Sigma}(x)$

Orbit ratio of TLM to linear variances, and surrogate :

$$\rho_o(x) := \max\Big(\frac{\sum_{i\in\mathcal{O}_o}\widehat{\Sigma}_{ii}(x)}{\sum_{i\in\mathcal{O}_o}\bar{\Sigma}^r_i},\ 0.05\Big), \qquad \boxed{\bar{\Sigma}_i(x) := \bar{\Sigma}^z_i\ \rho_{o(i)}(x)}$$

$\bar{\Sigma}^r := \operatorname{diag}\Sigma^{\mathrm{lin}}$ (orbit-constant) ; $\bar{\Sigma}^z : \bar{\Sigma}^r$ corrected for dequantisation (next slide). ρ carries the x-dependence (a ratio of r-variances) ; $\bar{\Sigma}^z$ carries the scale seen by the model.

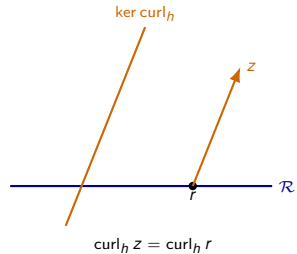
Why diagonal and orbit-wise : it is what each coefficient of the diffusion needs ; $n_{\mathrm{orb}} = 64$ numbers per state instead of d^2_ψ ; $\mathcal{G}$-invariant, like p_Δ.

Gaussian dequantisation

$B_{\ker} \in \mathbb{R}^{d_\psi \times n_{\ker}}$: basis of $\ker \operatorname{curl}_h$; $\mathrm{D} := \operatorname{diag} \bar{\Sigma}(x)$; $A_{\ker} := B_{\ker}^\top \mathrm{D}^{-1} B_{\ker}$.

$$z := r + B_{\ker} \mathrm{c}, \qquad \mathrm{c} \mid r \sim \mathcal{N}\big(-A_{\ker}^{-1} B_{\ker}^\top \mathrm{D}^{-1} r,\ A_{\ker}^{-1}\big)$$

Proposition. (i) $\operatorname{curl}_h z = \operatorname{curl}_h r : y$ is untouched. (ii) $\mathbb{R}^{d_\psi} = \mathcal{R} \oplus \ker \operatorname{curl}_h$; if $r = \Pi_{\mathcal{R}} w$, $w \sim \mathcal{N}(0, \mathrm{D})$ ($\Pi_{\mathcal{R}}$: oblique projector onto $\mathcal{R}$ along $\ker \operatorname{curl}_h$), then $z \sim \mathcal{N}(0, \mathrm{D})$. (iii) z has a density on $\mathbb{R}^{d_\psi}$.



Scale. $\bar{\Sigma}^z$ is the fixed point $\mathrm{D} = \bar{\Sigma}^r + \mathrm{pool}\big(\mathbb{E}\big[\mathrm{deq}_{\mathrm{D}}(r)^2\big] - \mathbb{E}\big[r^2\big]\big)$ under the Stokes+drag law : the orbit-pooled variance of z equals the reference. Computed (dyadic blocks). For $\rho \neq 1$ a mismatch $\approx 3\% \times$(spread of ρ) is left to the network.

Used in training (fresh c at every step : an augmentation) and to certify the sampler. The hypothesis of (ii) fails on real data (e^{tL} is not wavelet-diagonal) : the network still has work to do. Downstream everything is written for z ; r remains the physical residual.

Forward process : scale-adaptive Ornstein–Uhlenbeck

Per coefficient i of level $j = j(i)$, pseudo-time τ (no physical meaning) :

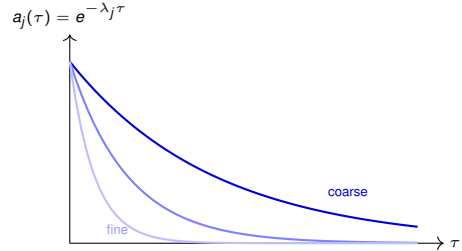
$$dR_{\tau,i} = -\lambda_j R_{\tau,i}\, d\tau + \sqrt{2\lambda_j}\, \sigma_i\, d\mathcal{W}_{\tau,i}, \quad R_0 = z, \quad \lambda_j = \lambda_0 2^{\beta j}.$$

Closed form.

$$R_\tau = a\, z + \sqrt{s}\, \varepsilon, \qquad a = e^{-\lambda_j \tau}, \quad s = \sigma^2(1 - a^2),$$

$\varepsilon \sim \mathcal{N}(0, I)$ independent of (x, z) ; $\sigma_i^2 := \mathbb{E}[z_i^2]$ (orbit-pooled, training pairs).

- training data are free : one ε per sample ;
- R_τ has a smooth positive density on $\mathbb{R}^{d_\psi}$ for $\tau > 0$;
- $R_\tau \to \mathcal{N}(0, \mathrm{diag}\, \sigma^2)$: z is forgotten ;
- fine scales (λ_j large) are blurred first.



Two independent sources of randomness : the physical noise ω_k (in $\mathcal{V}_h^0$, clock t) produces r ; the artificial noise ε (in $\mathbb{R}^{d_\psi}$, clock τ) only builds the generative model.

Score = denoiser, and denoisers are regressions

Conditional score $s(R, \tau; x) := \nabla_R \log p_\tau(R \mid x)$, with $p_\tau(R \mid x) = \int \mathcal{N}(R;\ az,\ \operatorname{diag} s)\, p(z \mid x)\, dz$.

Proposition (Tweedie). For every $\tau > 0$ (componentwise) :

$$s(R_\tau, \tau; x) = -\frac{R_\tau - a\,\mathbb{E}[z \mid R_\tau, x]}{s} = -\frac{\mathbb{E}[\varepsilon \mid R_\tau, x]}{\sqrt{s}}.$$

Proposition (regression). For any square-integrable target Y built from $(x, z, \varepsilon, \tau)$,

$$\arg\min_f\ \mathbb{E}\big[\|f(R_\tau, \tau, x) - \mathrm{Y}\|^2\big] = \mathbb{E}[\mathrm{Y} \mid R_\tau, \tau, x].$$

A conditional expectation is learned from samples of $(x, z, \varepsilon, \tau)$ alone, without ever evaluating p : denoising score matching (Vincent 2011). Next : the choice of Y that makes the Gaussian reference and the target of unit scale.

Gaussian reference and preconditioning

Reference law $p_{\mathrm{ref}}(z \mid x) := \mathcal{N}\big(0, \operatorname{diag} \bar{\Sigma}(x)\big)$. Per coefficient :

$$S_g = a^2 \bar{\Sigma} + s, \quad \mathrm{SNR} = \frac{a^2 \bar{\Sigma}}{s} = \frac{\kappa}{e^{2\lambda_j \tau} - 1}, \quad \kappa = \frac{\bar{\Sigma}}{\sigma^2}$$

Posterior under p_{ref}. $R_\tau \mid x \sim \mathcal{N}(0, S_g)$; $\mathbb{E}\big[z \mid R_\tau, x\big] = \frac{a\bar{\Sigma}}{S_g} R_\tau$, $\mathrm{Var}(z \mid R_\tau, x) = \frac{\bar{\Sigma} s}{S_g}$, score $-R_\tau / S_g$.

Diffusion time window. SNR decreases from $+\infty$ to 0. With κ_i^{lo}, κ_i^{hi} the 0.5% and 99.5% data percentiles,

$$\tau_{\min} = \min_i \frac{\log(1 + \kappa_i^{\mathrm{lo}} / \mathrm{SNR}_{\mathrm{hi}})}{2\lambda_{j(i)}}, \qquad T = \max_i \frac{\log(1 + \kappa_i^{\mathrm{hi}} / \mathrm{SNR}_{\mathrm{lo}})}{2\lambda_{j(i)}},$$

so every coefficient has $\mathrm{SNR} \geq \mathrm{SNR}_{\mathrm{hi}} = 10^2$ at $\tau_{\min}$ and $\mathrm{SNR} \leq \mathrm{SNR}_{\mathrm{lo}} = 10^{-3}$ at T. Replacing the law of R_T by $\mathcal{N}(0, S_g(T))$ costs a relative variance error $\leq \mathrm{SNR}_{\mathrm{lo}}$.

κ compares the conditional (this x) with the unconditional (data, all x) variance. Through $\rho(x)$ the SNR schedule adapts to the current flow.

The regression target is a rotation

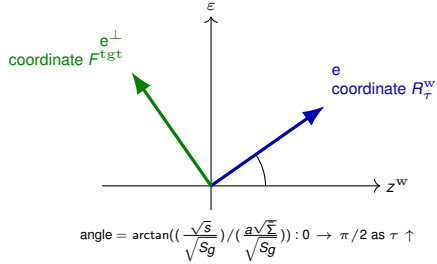
Whitened variables $z^{\mathrm{w}} := z/\sqrt{\Sigma}$, $R_\tau^{\mathrm{w}} := R_\tau/\sqrt{S_g}$:

$$R_\tau^{\mathrm{w}} = \left(\frac{a\sqrt{\Sigma}}{\sqrt{S_g}}\right) z^{\mathrm{w}} + \left(\frac{\sqrt{s}}{\sqrt{S_g}}\right) \varepsilon, \qquad \left(\frac{a\sqrt{\Sigma}}{\sqrt{S_g}}\right)^2 + \left(\frac{\sqrt{s}}{\sqrt{S_g}}\right)^2 = 1,$$

so $\mathrm{e} = \left(\left(\frac{a\sqrt{\Sigma}}{\sqrt{S_g}}\right), \left(\frac{\sqrt{s}}{\sqrt{S_g}}\right)\right)$, $\mathrm{e}^{\perp} = \left(-\left(\frac{\sqrt{s}}{\sqrt{S_g}}\right), \left(\frac{a\sqrt{\Sigma}}{\sqrt{S_g}}\right)\right)$ is an orthonormal basis of the $(z^{\mathrm{w}}, \varepsilon)$ plane.

Definition (target). the other coordinate,

$$F^{\mathrm{tgt}} := -\left(\frac{\sqrt{s}}{\sqrt{S_g}}\right) z^{\mathrm{w}} + \left(\frac{a\sqrt{\Sigma}}{\sqrt{S_g}}\right) \varepsilon$$



Lemma (every realisation, any law).

$$z^{\mathrm{w}} = \left(\frac{a\sqrt{\Sigma}}{\sqrt{S_g}}\right) R_\tau^{\mathrm{w}} - \left(\frac{\sqrt{s}}{\sqrt{S_g}}\right) F^{\mathrm{tgt}}, \qquad \varepsilon = \left(\frac{\sqrt{s}}{\sqrt{S_g}}\right) R_\tau^{\mathrm{w}} + \left(\frac{a\sqrt{\Sigma}}{\sqrt{S_g}}\right) F^{\mathrm{tgt}}.$$

The network sees R_τ ; F^{tgt} is the complementary information it must guess. This is the v-parameterisation (Salimans–Ho 2022).

Score decomposition : computed reference + learned correction

$F^\star(R_\tau, \tau, x) := \mathbb{E}\big[F^{\mathrm{tgt}} \mid R_\tau, \tau, x\big]$ for the true law $p(z \mid x)$.

Theorem. For every law $p(z \mid x)$ (no Gaussian assumption) and every $\tau > 0$:

$$\mathfrak{s}(R_\tau, \tau; x) = \underbrace{-\frac{R_\tau}{S_g}}_{\text{reference (computed)}} \underbrace{-\frac{\left(\frac{a\sqrt{\bar{\Sigma}}}{\sqrt{S_g}}\right)}{\sqrt{s}} F^\star}_{\text{correction (learned)}}, \qquad \mathbb{E}\big[z \mid R_\tau, x\big] = \frac{a\bar{\Sigma}}{S_g} R_\tau - \sqrt{\frac{s\bar{\Sigma}}{S_g}}\, F^\star.$$

Proof : condition the rotation lemma on R_τ, insert in Tweedie, use $\left(\frac{\sqrt{s}}{\sqrt{S_g}}\right)/(\sqrt{s}\sqrt{S_g}) = 1/S_g$.

Corollary 1 (fixed point). If $z \mid x \sim p_{\mathrm{ref}}$, then $F^\star \equiv 0$. $(z^{\mathrm{w}}, \varepsilon)$ is a standard Gaussian pair, a rotation keeps it standard, so F^{tgt} is independent of R_τ^{w}. False for r itself (supported on $\mathcal{R}$, while p_{ref} has full support) : this is why the model is trained on z.

Corollary 2. Under p_{ref}, $F^{\mathrm{tgt}} \sim \mathcal{N}(0, 1)$: unit target at every level and every τ, and the loss of $F_\theta \equiv 0$ equals 1.

$F_\theta \equiv 0$ *is* the TLM–Gaussian emulator. The network starts there and learns only $F^\star$, the departure of $p(z \mid x)$ from p_{ref}.

What the network has to learn

Proposition (first order). If $p(z \mid x) = \mathcal{N}(\mathrm{m},\ \mathrm{diag}\,\bar{\Sigma} + \Xi)$, with $\mathrm{m}^{\mathrm{w}} = \bar{\Sigma}^{-1/2}\mathrm{m}$, $\Xi^{\mathrm{w}} = \bar{\Sigma}^{-1/2}\,\Xi\,\bar{\Sigma}^{-1/2}$:

$$F^{\star} = -\Big(\frac{\sqrt{s}}{\sqrt{S_g}}\Big)\Big(\Xi^{\mathrm{w}}\Big(\frac{a\sqrt{\bar{\Sigma}}}{\sqrt{S_g}}\Big)R_\tau^{\mathrm{w}} + \mathrm{m}^{\mathrm{w}}\Big) + O\big(\|\Xi^{\mathrm{w}}\|^2 + \|\Xi^{\mathrm{w}}\|\,\|\mathrm{m}^{\mathrm{w}}\|\big).$$

Departure of p from p_{ref}	Taylor order	Enters $F^{\star}$ via	Relative size
off-diagonal $\Sigma(x)$, incl. cross-scale	1	Ξ^{w} off-diagonal	$O(1)$, grows with $k_f u_{\mathrm{rms}} \Delta$
localisation inside an orbit	1	Ξ^{w} diagonal	$O(1)$
mean shift $\mathbb{E}[z \mid x]$	2	m^{w}	$O(\eta_Q^{1/2})$
second-order variance	2	Ξ^{w}	$O(\eta_Q)$
skewness, higher cumulants	≥ 2	nonlinear in R_τ^{w}	$O(\eta_Q^{1/2})$

Three hypotheses. H_0 : Gaussian with x-independent covariance ; H_1 : $\mathcal{N}(0, \Sigma(x))$; H_2 : orders ≥ 2 present.

A validation loss can only reject H_0, against the best x-independent linear predictor (the *empirical linear floor*). H_1 versus H_2 is decided at fixed x, with solver ensembles.

Training

$$\mathcal{J}(\theta) = \mathbb{E}\big[\frac{1}{N_{\mathrm{lev}}}\sum_j \frac{1}{|\mathcal{I}_j|}\sum_{i\in\mathcal{I}_j} w_i(\tau)\big(F_{\theta,i} - F_i^{\mathrm{tgt}}\big)^2\big], \qquad N_{\mathrm{lev}} = J - j_{\min} + 1.$$

Minimiser $F^\star$ for any weight $w > 0$. Default $w \equiv 1$: equivalently, the ε-error weighted by $1 + \mathrm{SNR}^{-1}$, or the whitened z-error weighted by $1 + \mathrm{SNR}$.

	One training sample	Source
1	(x, y, m, ρ) from the dataset ; $r = \mathrm{curl}_h^+(y - m)$	solvers, TLM
2	$\bar{\Sigma} = \bar{\Sigma}^z \rho_{o(\cdot)}$; $z = r + B_{\mathrm{ker}}$c, fresh c	reference ; dequantisation
3	random dyadic shift of z, $\mathrm{curl}_h^+ m$, $\mathrm{curl}_h^+ x$	symmetry $\mathcal{G}$
4	$\tau \sim \log \mathcal{U}[\tau_{\min}, T]$; $R_\tau = az + \sqrt{s}\,\varepsilon$	forward OU
5	$F^{\mathrm{tgt}} = -(\frac{\sqrt{s}}{\sqrt{S_g}})z/\sqrt{\bar{\Sigma}} + (\frac{a\sqrt{\bar{\Sigma}}}{\sqrt{S_g}})\varepsilon$	rotation
6	$F_\theta\,(R_\tau/\sqrt{S_g},\ \mathrm{curl}_h^+ m/\bar{m},\ \mathrm{curl}_h^+ x/\bar{x},\ \mathrm{l})$; AdamW step	network

- every input and the target have unit scale under p_{ref}, at every level and every τ ;
- last layer zero-initialised : step 0 *is* the reference ; gains are measured against the paired $F_\theta \equiv 0$ baseline on held-out trajectories.

AdamW, warm-up + cosine, clipping 1, EMA. $\bar{m}_l$, $\bar{x}_l$: orbit-pooled RMS of $\mathrm{curl}_h^+ m$, $\mathrm{curl}_h^+ x$ (training pairs). l_j : level mean of $\log \mathrm{SNR}$.

Reverse step : mean from the network, variance computed

Grid $T = \tau_N > \cdots > \tau_1 = \tau_{\min} > \tau_0 = 0$. For $\tau > \tau'$, :

$$p(R_{\tau'} \mid R_\tau, x) = \int \underbrace{p(R_{\tau'} \mid R_\tau, z)}_{\text{Gaussian bridge}} \underbrace{p(z \mid R_\tau, x)}_{\text{posterior}} \, dz, \qquad R_{\tau'} \mid R_\tau, z \sim \mathcal{N}\big(b_1 R_\tau + b_2 z,\ v_{\mathrm{br}}\big),$$

$b_1 = \dfrac{a_{\tau|\tau'} s_{\tau'}}{s_\tau}$, $b_2 = \dfrac{a_{\tau'} s_{\tau|\tau'}}{s_\tau}$, $v_{\mathrm{br}} = \dfrac{s_{\tau'} s_{\tau|\tau'}}{s_\tau}$, $a_{\tau|\tau'} = a_\tau / a_{\tau'}$, $s_{\tau|\tau'} = \sigma^2(1 - a^2_{\tau|\tau'})$.

Sampler step. Gaussian with the first two moments of the transition, posterior mean from the network, posterior variance from p_{ref} :

$$R_{\tau'} = b_1 R_\tau + b_2 \hat{z} + \sqrt{v_{\mathrm{br}} + b_2^2 \, \bar{\Sigma} s_\tau / S_g(\tau)}\, \varepsilon', \qquad \hat{z} = \frac{a_\tau \bar{\Sigma}}{S_g(\tau)} R_\tau - \sqrt{\frac{s_\tau \bar{\Sigma}}{S_g(\tau)}}\, F_\theta .$$

Corollary (what is exact).

1. if $p = p_{\mathrm{ref}}$ and $F_\theta \equiv 0$: the chain returns $z \sim p_{\mathrm{ref}}$ **, for any number of steps** ;
2. otherwise : mean error $\propto F_\theta - F^\star$; the posterior-variance error $b_2^2 \big(\mathrm{Var}(z \mid R_\tau, x) - \bar{\Sigma} s_\tau / S_g\big)$ is $O(\max \Delta\tau)$ on interior steps ; the final jump $\tau_{\min} \to 0$ ($b_2 = 1$) costs $\leq \bar{\Sigma}/(1 + \mathrm{SNR}_{\mathrm{hi}})$, i.e. $\leq 1\%$ of $\bar{\Sigma}$.

The emulator, one step

```
# one emulator step:  y ~ p_theta(. | x)
m     = Phi_Delta(x)               # noiseless solver
rho   = rho_fn(x)                  # TLM ratio, fixed probes
Sb    = Sbar_z * rho[orbit]        # reference
cm,cx = curl_pinv(m)/mbar, curl_pinv(x)/xbar
R     = sqrt(Sb*a(T)**2 + s(T)) * normal()   # prior
for tau, tau_p in grid(T -> tau_min -> 0):
    a, s         = ou(tau)
    Sg           = a**2 * Sb + s
    b1, b2, vbr = bridge(tau, tau_p)
    F    = F_theta(R/sqrt(Sg), cm, cx, lsnr(tau))
    zhat = a*Sb/Sg*R - sqrt(s*Sb/Sg)*F   # decomposition
    var  = vbr + b2**2 * Sb*s/Sg         # total variance
    R    = b1*R + b2*zhat + sqrt(var)*normal()
y = m + curl(R)          # R = z at tau = 0; curl kills B_ker c
```

Rollout : $x^{(\iota+1)} \sim p_\theta(\cdot \mid x^{(\iota)})$.

Properties

- no projection, no physical noise drawn ;
- every y is divergence free and zero mean ;
- ancestral sampling with the bridge.

Cost of one lag

- solver lag : $n_{\mathrm{s}} = 100$ micro-steps ;
- TLM : $\approx 2 n_{\mathrm{probe}} n_{\mathrm{s}}$ step-equivalents ;
- $N_\tau = 64$ network calls, two FFT curls.

Production configuration and flow regime

$J = 7$ (128^2), $j_{\min} = 4$, Bubnov, IF-RK4 $dt = 10^{-3}$, $n_{\mathrm{s}} = 100$, $\Delta = 0.1$; $\nu = 10^{-4}$, $\mu_d = 0.5$, $\varepsilon_{\mathrm{inj}} = 0.05$, $\alpha_F = 5$; $n_{\mathrm{probe}} = 8$.

Quantity	Definition	Value	Reading
forcing wavenumber	$k_f^2 = \eta_{\mathrm{inj}}/\varepsilon_{\mathrm{inj}}$ (exact)	66.2	94% of $\varepsilon_{\mathrm{inj}}$ in c_A
Kraichnan scale	$(\eta_{\mathrm{inj}}/\nu^3)^{1/6}$	$246 = 0.61\, k_{\max}$	resolved
enstrophy above $n/4$	measured	4.4% (tol. 5%)	at resolution limit
Reynolds number	$Re_f = u_{\mathrm{rms}}/(\nu k_f)$	≈ 40	moderately nonlinear
turnover time at k_f	$1/(k_f u_{\mathrm{rms}})$	0.057	$\Delta = 1.7$ turnovers
friction scale	$\ell_{\mathrm{fr}} = (\varepsilon_{\mathrm{inj}}/\mu_d^3)^{1/2}$	0.63	$\approx$ box : condensate risk
drag share of dissipation	$2\mu_d \langle E \rangle / \varepsilon_{\mathrm{inj}}$	0.69 (linear : 0.55)	inverse transfer
box-scale energy fraction	$E_{\lvert \boldsymbol{k} \rvert = 2\pi} / \langle E \rangle$		no condensate iff small

Consequences. (i) Δ spans 1.7 turnovers : transport of the noise by x is $O(1)$, an x-independent Gaussian cannot hold at fine scales. (ii) Higher Re requires $J = 8$. (iii) If the box-scale fraction is not small, raise μ_d ($\ell_{\mathrm{fr}} \leq 1/4 \iff \mu_d \geq (16\varepsilon_{\mathrm{inj}})^{1/3} \approx 0.93$).

Certificates : every component against a known answer

Certificate (what it would falsify)	Known answer	Measured, $J = 7$
connection coefficients vs. quadrature	0	$\leq 5.5 \times 10^{-7}$
advective energy neutrality, rough fields	0	1.1×10^{-14}
$\|\|Cx\|\|$; P $\mathrm{curl}_h = \mathrm{curl}_h$; $\mathrm{curl}_h\,\mathrm{curl}_h^+ = \mathrm{id}$ on $\mathcal{V}_h^0$	0	2×10^{-16} ; 4.4×10^{-16} ; 8.1×10^{-15}
Richardson order of IF-RK4	4	4.00
stochastic Taylor–Green : OU on an invariant shell (integrator)	$\max z < 5.25$	2.02 ; KS $p = 0.97$
dequantiser : $\mathrm{curl}_h\, B_{\ker} = 0$; orbit Var /D $- 1$ (fixed point $\bar{\Sigma}^z$)	0 ; ≤ 0.006	4.7×10^{-6} ; ≤ 0.003
sampler with $F_\theta \equiv 0$, 64 steps (bridge + total variance)	≤ 0.022	0.020
pathwise energy budget, residual/bound	< 1	0.07
stationary identity, solver data	0	$+0.22\% \pm 0.19\%$ (1.2σ)

The scheme, the stochastic integrator, the dequantiser and the sampler are each validated separately, before any learning is involved.

Summary

1. **scheme.** Wavelet–Galerkin with connection coefficients : energy-neutral advection, G_2-orthogonal Leray projector, commuting operators (no splitting error), IF-RK4.
2. **Well-posed target.** The forced, dragged chain has a unique invariant law and a smooth positive transition density on $\mathcal{V}_h^0$: the score exists.
3. **Unconstrained generative coordinates.** The discrete stream function plus dequantisation : samples are divergence free by construction.
4. **Reference.** The residual is a polynomial of the injected Gaussian noise ; its Taylor-linear law is $\mathcal{N}(0, \Sigma(x))$, computed by tangent-linear probes. At $J = 7$ it matches the solver to $\approx 10\%$, where the x-independent law is off by $\approx 16\times$.
5. **score decomposition.** For any law : score $=$ Gaussian reference $+$ learned correction $F^\star$, with $F^\star \equiv 0$ for the reference and unit-scale targets.
6. **sampler for the reference** at any step count ; physics checked with no free constant.

Not an accelerator : a calibrated, differentiable conditional law $p_\theta(y \mid x)$ with a usable score, which the solver does not provide : likelihoods, data assimilation, rare-event tilting, ensembles.

Limitations and outlook

1. **What the loss proves.** It rejects H_0 only ; content beyond Taylor order 1 needs the fixed-x panel (emulator mean shift against its null, lattice term separated).
2. **Emulator defects.** -5% conditional energy budget over one drag time ; stationary rollout missing.
3. **Diagonal surrogate.** Off-diagonal and within-orbit structure of $\Sigma(x)$ are $O(1)$ and left to the network. Next : low-rank $+$ diagonal $\Sigma(x)$ from the probes already computed.
4. **Cost.** $10.6\times$ the solver, $\approx$ 90% TLM. Next : amortise $\rho(x)$ with a small regressor ; raise dt (CFL 0.384 leaves a factor ≈ 7).
5. **Mean shift** computable by a second-order tangent model : would remove $\mathbf{m}^{\mathrm{W}}$ from $F^\star$.
6. **Loss weighting.** Equal weight per level although $|\mathcal{I}_j| = 3 \cdot 4^j$: defensible, but the headline gain depends on it.
7. **Discrete vs continuum.** Ergodicity is proved for the SDE and for the chain ; $\pi_\infty^{dt} \to \pi_\infty$ is not.
8. **Scope.** Periodic domain : bounded domains lose circulants, commutation, the dyadic group and the Fourier stream function.

Outlook : a cheap drift is a different method. With a coarse Φ_Δ^{c}, $\mathscr{R}_x(0) \neq 0$ (deterministic model error) : the parity argument fails and the reference must acquire a mean.

Backup : notation

Symbol	Meaning	Symbol	Meaning
$n = 2^J,\ h$	grid size, mesh	$\Phi,\ \Phi_\Delta$	micro-step, noiseless lag map
$W_{\mathrm{fwd}}, W_{\mathrm{inv}}, \mathbb{W}$	analysis, synthesis, 2D change	$\omega_k,\ \boldsymbol{\omega}$	injected noise, one lag
M, D, K, G	mass, derivative, 2nd deriv., Gram	Q_ψ, B_F, Q	forcing : covariance, factor, $B_F B_F^\top$
$G_2, G_{2,w}$	L^2 metric (scaling / wavelet)	$\widetilde{Q},\ \mathsf{f}$	injected covariance, EM weight
$C,\ \mathcal{K}$	weak divergence, weak Laplacian	$E, Z_{\mathrm{gal}}, \varepsilon_{\mathrm{inj}}$	energy, dissipation, injection
$M^{-1}K$	$M^{-1}K$	$m,\ r,\ z$	prediction, residual, dequantised
$\mathsf{P},\ \mathcal{F}$	Leray projector, Nyquist filter	$B_{\ker},\ \mathsf{c}$	$\ker \mathrm{curl}_h$ basis, coefficients
$L, \mathcal{H}, \mathcal{N}$	linear generator, advection	$\Sigma(x),\ \Sigma^{\mathrm{lin}}$	reference matrix, Stokes+drag
$\mathcal{V}_h,\ \mathcal{V}_h^0$	$\ker C \cap \mathrm{ran}\,\mathcal{F}$, zero-mean part	$\rho_o,\ \bar{\Sigma}$	TLM ratio, surrogate
$\mathrm{curl}_h, \mathrm{curl}_h^+, \mathcal{R}$	curl, pseudo-inverse, $\mathrm{ran}\,\mathrm{curl}_h^+$	$\bar{\Sigma}^r,\ \bar{\Sigma}^z$	linear diag., dequantised fixed point
$j(i), o(i), \mathcal{G}$	level, orbit, dyadic group	$a, s, \sigma^2, \lambda_j$	OU decay, variance, stat. var., rate
η_Q	forcing amplitude (analysis)	$S_g, \mathrm{SNR}, \kappa$	$a^2\bar{\Sigma} + s,\ a^2\bar{\Sigma}/s,\ \bar{\Sigma}/\sigma^2$
p_{ref}	$\mathcal{N}(0, \mathrm{diag}\,\bar{\Sigma})$	$(\frac{a\sqrt{\bar{\Sigma}}}{\sqrt{S_g}}), (\frac{\sqrt{s}}{\sqrt{S_g}})$	$a\sqrt{\bar{\Sigma}/S_g},\ \sqrt{s/S_g}$
$\mathcal{B},\ \mathcal{W}$	physical / diffusion Brownian	$F^{\mathrm{tgt}}, F^\star, F_\theta$	target, its cond. mean, network
$\boldsymbol{\xi}$	spatial variable	$b_1, b_2, v_{\mathrm{br}}$	bridge coefficients

Backup : wavelet–Galerkin proofs

Two-term coefficients. Test ϕ_t (filter $\mathfrak{h}^t$), trial φ (filter $\mathfrak{h}$), $\sum_i \mathfrak{h}_i = 2$. Insert both refinement equations, substitute $\xi \mapsto 2\xi$ ($\varphi^{(q)}$ gains 2^q, the Jacobian $\frac{1}{2}$) :

$$\gamma_l^{(q)} = 2^{q-1} \sum_{i,j} \mathfrak{h}_i^t \mathfrak{h}_j \, \gamma_{2l+j-i}^{(q)} \iff \mathcal{T}\gamma^{(q)} = 2^{-q}\gamma^{(q)}, \qquad \mathcal{T}_{l,l'} = \tfrac{1}{2} \sum_i \mathfrak{h}_i^t \mathfrak{h}_{i+l'-2l}.$$

Normalisation : $\sum_l l^q \varphi(\xi - l) = \xi^q +$ lower order ; differentiate q times and pair with ϕ_t ($\int \phi_t = 1$) : $\sum m_l = 1$, $\sum l\, d_l = 1$, $\sum l^2 k_l = 2$. Symmetry imposed (m, k even, d odd).

Triad coefficients. $\Gamma_{p,q} = \langle \phi_t, \varphi(\cdot - p)\varphi(\cdot - q)\rangle$, $\Lambda_{p,q} = \langle \phi_t, \varphi(\cdot - p)\varphi'(\cdot - q)\rangle$. With $\mathcal{T}^{(3)}_{(p,q),(p',q')} = \sum_i \mathfrak{h}_i^t \mathfrak{h}_{i+p'-2p} \mathfrak{h}_{i+q'-2q}$:
$\mathcal{T}^{(3)}\Gamma = 2\Gamma$, $\mathcal{T}^{(3)}\Lambda = \Lambda$, $\sum_{p,q} \Gamma_{p,q} = 1$, $\sum_q q\Lambda_{p,q} = m_p$. Certified against quadrature at every run ($\le 6 \times 10^{-7}$).

Energy neutrality. $\int (\boldsymbol{u}\cdot\nabla\boldsymbol{w})\cdot\boldsymbol{w} = \frac{1}{2}\int \boldsymbol{u}\cdot\nabla|\boldsymbol{w}|^2 = -\frac{1}{2}\int(\nabla\cdot\boldsymbol{u})|\boldsymbol{w}|^2$ cancels the Temam term, so $b(\boldsymbol{u},\boldsymbol{w},\boldsymbol{w}) = 0$. The stencils evaluate these integrals , and in Bubnov mode $\boldsymbol{w}_h = \sum_k W_k \varphi_k$ is a test function, so $\mathrm{w}^\top \mathcal{N}(\mathrm{u};\mathrm{w}) = h^{-2} b(\boldsymbol{u}_h, \boldsymbol{w}_h, \boldsymbol{w}_h) = 0$. □

Cost. The skew stencil $\Lambda + \frac{1}{2}\Lambda^\top$ and Γ are stored as SVDs of rank 10–12 (to 10^{-10}) : the 2D contraction is a sum of products of separable 17-tap circulant filterings, $O(R^2 \cdot 17 \cdot n^2)$; the n^3 tensor is never formed.

Bubnov vs Petrov. Petrov (test space $\widetilde{V}_J$) gives $M = I$ but $M_2 \neq G_2$: energy neutrality only $O(h^8)$ on smooth fields, 10^{-6}–10^{-4} on rough ones, and the Leray projector is no longer the pressure elimination. Validated, not used.

Initial data. Coefficients of an analytic field are not its samples : Bubnov multiplier $\widehat{\varphi}(\vartheta)e^{i\vec{c}\vartheta}\,/\,\widehat{g}(\vartheta)$. Using samples as coefficients is an $O(h^2)$ variational crime.

Backup : Leray projector, commutation, mean flow

P_0 **is the G_2-orthogonal projector onto** $\ker C$**.** KKT for $\min_{Cv=0} \|v-u^*\|^2_{G_2}$: $G_2(v-u^*)+C^\top \varpi = 0$, $Cv=0 \Rightarrow$ $\mathcal{S}\varpi = Cu^* \in \operatorname{ran} C = \operatorname{ran}\mathcal{S}$. Any generalised inverse $\mathcal{S}^g$ ($\mathcal{S}\mathcal{S}^g\mathcal{S}=\mathcal{S}$) gives a solution ; solutions differ by $\ker\mathcal{S} = \ker C^\top$, annihilated by $C^\top$, so $C^\top\mathcal{S}^gC$ is unique. Then $C\mathrm{P}_0 = C - \mathcal{S}\mathcal{S}^gC = 0$, $G_2\mathrm{P}_0 = G_2 - C^\top\mathcal{S}^gC$ is symmetric, $\mathrm{P}_0^2 = \mathrm{P}_0$. □

$\mathcal{S}^g$ **from two** $n\times n$ **eigenproblems.** $G_2^{-1} = G^{-1}\otimes G^{-1}$ gives $\mathcal{S} = A_p\otimes B_p + B_p\otimes A_p$, $A_p = DG^{-1}D^\top \succeq 0$, $B_p = MG^{-1}M^\top \succ 0$. With $A_pY = B_pY\,\mathrm{diag}(\theta)$, $Y^\top B_pY = I$: $(Y\otimes Y)^\top\mathcal{S}(Y\otimes Y) = \mathrm{diag}(\theta_i+\theta_j)$, hence

$$\mathcal{S}^g\,\mathrm{vec}(X) = \mathrm{vec}\Big(Y\Big[\frac{(Y^\top XY)_{ij}}{\theta_i+\theta_j}\,\mathbb{1}_{\theta_i+\theta_j>0}\Big]Y^\top\Big).$$

Commutation. Every block of P_0 lies in the commutative algebra generated by circulant $\otimes$ circulant ($\mathcal{S}^g$ included) ; $L = I_2\otimes L_1$ acts identically on both components ; so $\mathrm{P}L = L\mathrm{P}$, $\mathrm{P}e^{tL} = e^{tL}\mathrm{P}$, and $\mathcal{V}_h$ is invariant under the exact flow and every stage of the scheme. e_N is an eigenvector of every circulant, so Π_{Ny} commutes with everything and is G-orthogonal.

Mean flow. $\bar{u} = n^{-2}\mathbf{1}^\top u$ is the spatial mean ($\sum_k \varphi_{J,k}\equiv 1$). Bubnov, $M\mathbf{1} = \mathbf{1}:\mathbf{1}^\top M_2^{-1}\mathcal{N}(u) \propto b(\boldsymbol{u}_h,\boldsymbol{u}_h,\mathbf{e}_c) = -\frac{1}{2}\int(\nabla\cdot\boldsymbol{u}_h)u_{h,c} = 0$ (a combination of the entries of $Cu=0$) ; P does not change the mean. Hence $\dot{\bar{u}} = -\mu_d\bar{u}$ (measured drift 10^{-6}, float32).

Dimension of $\mathcal{V}_h$**.** On $\operatorname{ran}\mathcal{F}$: $2(n-1)^2$ directions ; the constraint is one equation per non-Nyquist mode except $(0,0)$ where C vanishes : $\dim\mathcal{V}_h = (n-1)^2+1$, and removing the two mean-flow directions gives $\dim\mathcal{V}_h^0 = (n-1)^2-1$.

Backup : stream function, forcing, dequantisation

(1) $C\,\mathrm{curl}_h = 0$. $C(U,V) = DUM^\top + MVD^\top$. With $U = \Psi(M^{-1}D)^\top$, $V = -(M^{-1}D)\Psi$ and $M(M^{-1}D) = D$ (M symmetric) : $D\Psi D^\top - D\Psi D^\top = 0$; $\mathcal{F}$ commutes with C.
(2)–(3) Range and kernel. Per Fourier mode, $\mathrm{a} = \hat{d}/\hat{m}$, $\hat{d} = i\mathfrak{d}$ with $\mathfrak{d}$ real, odd, $\mathfrak{d}(\vartheta) = 0 \iff \vartheta \in \{0, \pi\}$. C reads $\hat{m}_1\hat{m}_2(\mathrm{a}_1\hat{U} + \mathrm{a}_2\hat{V}) = 0$ and curl_h gives $(\hat{U}, \hat{V}) = (\mathrm{a}_2, -\mathrm{a}_1)\hat{\Psi}$, which spans the constraint line iff $(\mathrm{a}_1, \mathrm{a}_2) \neq 0$; off the Nyquist lines this fails only at $(0,0)$. Hence $\mathrm{ran}\,\mathrm{curl}_h = \mathcal{V}_h^0$ and $\ker\,\mathrm{curl}_h$ is spanned by the $(0,0)$ mode and the $2n-1$ Nyquist-line modes : $\dim\ker\mathrm{curl}_h = 2n$. Pseudo-inverse : $\hat{\Psi} = (\overline{\mathrm{a}_2}\hat{U} - \overline{\mathrm{a}_1}\hat{V})/(|\mathrm{a}_1|^2 + |\mathrm{a}_2|^2)$, 0 on Nyquist lines and at $(0,0)$.
(4) Bubnov. $\langle\varphi_a\varphi_b, \partial_{\xi_2}\psi_h\rangle = h^2(G\Psi D^\top)_{ab}$, so the L^2 projection has coefficients $G^{-1}(G\Psi D^\top)G^{-1} = \Psi(G^{-1}D)^\top = \Psi(M^{-1}D)^\top$ ($M = G$). Same for $-\partial_{\xi_1}\psi_h$.

Forcing. $Q_\psi^{1/2}$ is invertible, so $\mathrm{ran}\,B_F = \mathrm{ran}\,\mathrm{curl}_h = \mathcal{V}_h^0$ and $Q \succ 0$ on $\mathcal{V}_h^0$. Q_ψ is level-diagonal, hence invariant under level-preserving permutations ; curl_h commutes with translations : Q is $\mathcal{G}$-invariant. Invariant-measure regularity (viscous balance $q/(2\nu k^2)$) : $\mathbb{E}\big[\|\boldsymbol{u}\|_{H^\mathrm{s}}^2\big] < \infty \iff \mathrm{s} < \frac{1+\alpha_F}{2}$. Forcing in velocity coordinates instead : $\approx 9\%$ of the training loss is a purely linear-Gaussian correlation with no dynamical content.

Dequantisation. As a function of c, the log-density of $\mathcal{N}(0,\mathrm{D})$ at $r + B_{\ker}\mathrm{c}$ is $-\frac{1}{2}\left(\mathrm{c}^\top A_{\ker}\mathrm{c} + 2\mathrm{c}^\top B_{\ker}^\top \mathrm{D}^{-1}r\right) + \text{const}$: the conditional law of the $\ker\mathrm{curl}_h$ component of $w \sim \mathcal{N}(0,\mathrm{D})$ given $\Pi_{\mathcal{R}}\,w = r$. Since $\mathbb{R}^{d_\psi} = \mathcal{R} \oplus \ker\mathrm{curl}_h$ with constant Jacobian, $(r,\mathrm{c}) \mapsto z$ reconstructs w in law. □

Backup : time integration

IF-RK4 (Lawson 1967). $\mathcal{E}_{1/2} = e^{dtL/2}$, $\mathcal{E}_1 = e^{dtL}$, applied modally :

$$k_1 = \mathcal{H}(\mathrm{u}),\;\; k_2 = \mathcal{H}(\mathcal{E}_{1/2}(\mathrm{u} + \tfrac{dt}{2}k_1)),\;\; k_3 = \mathcal{H}(\mathcal{E}_{1/2}\mathrm{u} + \tfrac{dt}{2}k_2),\;\; k_4 = \mathcal{H}(\mathcal{E}_1\mathrm{u} + dt\,\mathcal{E}_{1/2}k_3),$$

$$\Phi(\mathrm{u}) = \mathrm{P}\big[\mathcal{E}_1(\mathrm{u} + \tfrac{dt}{6}k_1) + \tfrac{dt}{3}\mathcal{E}_{1/2}(k_2 + k_3) + \tfrac{dt}{6}k_4\big].$$

Stable on $i[-2\sqrt{2}, 2\sqrt{2}]$ with $\mathrm{CFL} = dt\,c_D \max(|U| + |V|)$; Heun and Euler are stable on no part of the imaginary axis.

Proposition 1. $\mathrm{f}(z)^2 = \int_0^1 e^{2zt}\,dt$. increment : $\mathrm{Cov}(\omega^{\mathrm{ex}})_{ll'} = dt\,\hat{Q}_{ll'}\,\mathrm{f}(\frac{z_l+z_{l'}}{2})^2$; scheme : $\mathrm{Cov}(\omega)_{ll'} = dt\,\hat{Q}_{ll'}\,\mathrm{f}(z_l)\mathrm{f}(z_{l'})$. $\varphi := \log \mathrm{f}$ is convex with $\varphi'' = 2\,\mathrm{Var}_{\mathrm{tilt}}(t) \le \frac{1}{6}$ (t exponentially tilted $\mathcal{U}[0,1]$, variance $\le \frac{1}{12}$). Hence $0 \le \varphi(z_l) + \varphi(z_{l'}) - 2\varphi(\frac{z_l+z_{l'}}{2}) \le \frac{1}{4}(z_l - z_{l'})^2 \sup \varphi''$. □

Proposition 2. Stationarity $\hat{C} = e^{dt\Lambda}\hat{C}e^{dt\Lambda} + \mathrm{Cov}(\omega)$ gives $\hat{C}^{dt}_{ll'} = \mathrm{Cov}(\omega)_{ll'}/(1 - e^{z_l+z_{l'}})$; use $dt\,\mathrm{f}(\frac{w}{2})^2/(1 - e^{w}) = dt/|w|$, $w = z_l + z_{l'}$. E and Z_{gal} use only the modal diagonal ($\Upsilon^\top M\Upsilon = I$, $\Upsilon^\top K\Upsilon = \mathrm{diag}\,\ell$). □

increment if dt is raised. With Gauss–Legendre $(t_m, w_m)_{m \le n_{\mathrm{GL}}}$ on $[0,1]$: $\omega_k = \sqrt{dt}\sum_m \sqrt{w_m}\,e^{t_m\,dt\,L}B_F\zeta_{k,m}$, covariance up to $\lesssim z_{\max}^4/270$ ($n_{\mathrm{GL}} = 2$).

Why IF-RK4 and not exponential Euler. $r = \mathrm{curl}_h^+(y - m)$ is the response of the scheme actually used, whatever its order : the drift order does not bias learning. But Lawson–Euler applies forward Euler to skew advection ; per mode $|e^{-dt\nu k^2}(1 + i\,dt\,ku)| \le 1$ holds if $dt \le 2\nu/u_{\max}^2$, i.e. $u_{\max} \le 0.45$ here, violated in production. IF-RK4 at $\mathrm{CFL} = 0.384$ leaves a factor ≈ 7 in dt, which divides the dominant TLM cost $2n_{\mathrm{probe}}n_{\mathrm{s}}$.

What exponential Euler–Maruyama gets right

Modal basis of L ($K\Upsilon = M\Upsilon\,\mathrm{diag}\,\ell$, $\Upsilon^\top M\Upsilon = I$), $I = (i,j)$: $z_I := dt\,(\nu(\ell_i + \ell_j) - \mu_d) \le -\mu_d\,dt < 0$.

Proposition 1 (one increment). With $\omega_k^{\mathrm{ex}} = \int_{t_k}^{t_{k+1}} e^{(t_{k+1}-t')L} B_F\, d\mathcal{B}_{t'}$,

$$\mathrm{Cov}(\omega_k) = \mathrm{Cov}(\omega_k^{\mathrm{ex}}) \circ \mathrm{R} \ \text{(Hadamard)}, \qquad \mathrm{R}_{II} = 1, \qquad 0 \le \log \mathrm{R}_{II'} \le \tfrac{1}{24}(z_I - z_{I'})^2.$$

Modal variances are exact ; cross-mode covariances are inflated by at most $e^{z_{\max}^2/24} \approx 1 + 5\cdot 10^{-5}$ ($z_{\max} \approx 0.033$). Not in general : Q_ψ is wavelet-diagonal, L is modal-diagonal.

Proposition 2 (linear energetics). If $\mathcal{H} = 0$, the stationary modal covariance of the chain is $\hat{C}^{dt} = \hat{C} \circ \mathrm{R}$ with $\hat{C}_{II'} = dt\,\hat{Q}_{II'}/|z_I + z_{I'}|$, the SDE's. Hence $\langle E \rangle$ and $\langle Z_{\mathrm{gal}} \rangle$, which involve modal variances only, are **for every** dt. Unweighted noise (f $\equiv 1$) : relative error $\approx |z_I|$ per mode.

The learning problem is unaffected either way : $\Sigma(x)$ is built from $dt\,\widetilde{Q}$, the noise actually injected.

Well-posedness : the learning target exists

Proposition A (reduced SDE on $\mathcal{V}_h^0$). Global strong solution ; unique invariant law π_∞ with a smooth positive density. Lyapunov : $\mathscr{L}E \le -2\mu_d E + \varepsilon_{\mathrm{inj}}$ ($Z_{\mathrm{gal}} \ge 0$, advection neutral) ; additive noise non-degenerate on $\mathcal{V}_h^0$.

Proposition B (the discrete chain : the object actually learned). With an analysis-only energy limiter ($\Phi \to \mathrm{P}e^{dt\mathcal{L}}$ above $E_{\max} = 20\,\varepsilon_{\mathrm{inj}}/2\mu_d$) : drift condition + minorisation on sublevel sets $\Rightarrow$ (Harris) unique π_∞^{dt}, geometric ergodicity, smooth positive density. Needed because Φ is an explicit polynomial of degree 16 : explicit schemes with super-linear drift can have divergent moments (Hutzenthaler–Jentzen–Kloeden 2011).

Corollary (a score exists). $y = \Phi(\mathrm{u}_{n_\mathrm{s}-1}) + \omega_{n_\mathrm{s}-1}$ with $\omega_{n_\mathrm{s}-1}$ independent of $\mathrm{u}_{n_\mathrm{s}-1}$ and $dt\,\widetilde{Q} \succ 0$ on $\mathcal{V}_h^0$: $p_\Delta(\cdot \mid x)$ is a Gaussian convolution, smooth and positive on $\mathcal{V}_h^0$, so $\nabla \log p_\Delta(\cdot \mid x)$ exists.

On stored trajectories the limiter is inactive : $\max E = 0.045 \ll E_{\max} = 1$ (snapshots). Not claimed : $\pi_\infty^{dt} \to \pi_\infty$ as $dt \to 0$, nor convergence as $h \to 0$.

Backup : ergodicity of the SDE and of the chain

SDE. Itô with $G_2\mathrm{P} = \mathrm{P}^\top G_2$, $\mathrm{P}\mathrm{u} = \mathrm{u}$, $\mathrm{u}^\top \mathscr{N}(\mathrm{u}) = 0$: $\mathscr{L}E = -2\mu_d E - 2\nu Z_{\mathrm{gal}} + \varepsilon_{\mathrm{inj}} \le -2\mu_d E + \varepsilon_{\mathrm{inj}}$. (1) Locally Lipschitz drift + Lyapunov function : global strong solution (Khasminskii). (2) $\mathbb{E}\big[E(t)\big] \le e^{-2\mu_d t}E(0) + \varepsilon_{\mathrm{inj}}/(2\mu_d)$: tightness, existence by Krylov–Bogoliubov. (3) Additive noise non-degenerate on the finite-dimensional $\mathcal{V}_h^0$: strong Feller and irreducible, uniqueness (Doob–Khasminskii) ; smooth positive density by elliptic regularity of $\mathscr{L}^*\pi = 0$ (Bogachev–Krylov–Röckner–Shaposhnikov 2015). Remark : the discrete Poincaré inequality $Z_{\mathrm{gal}} \ge |\ell_1|E$ on $\mathcal{V}_h^0$ would give existence already at $\mu_d = 0$; drag is a modelling choice (large-scale level, no slow condensate mode).
Chain. $\tilde{\Phi} = \Phi$ if $E \le E_{\max}$, $\mathrm{P}e^{dtL}$ otherwise ; $\mathrm{u}_{k+1} = \tilde{\Phi}(\mathrm{u}_k) + \omega_k$. Let $\delta := \sup_{E(\mathrm{u})\le E_{\max}}[E(\Phi(\mathrm{u})) - e^{-2\mu_d dt}E(\mathrm{u})]_+/dt$ (finite ; $O(dt^4)$ by the order of IF-RK4 and the ODE's energy law).

1. *Drift.* Above $E_{\max}$: P is G_2-contractive and e^{dtL} has symbol e^z, $z \le -\mu_d dt$, so $E(\tilde{\Phi}(\mathrm{u})) \le e^{-2\mu_d dt}E(\mathrm{u})$; below : definition of δ. The cross term is centred and $\mathbb{E}\big[E(\omega_k)\big] \le dt\,\varepsilon_{\mathrm{inj}}$ ($0 < \mathrm{f} \le 1$). Hence $\mathbb{E}\big[E(\mathrm{u}_{k+1}) \mid \mathrm{u}_k\big] \le e^{-2\mu_d dt}E(\mathrm{u}_k) + dt(\varepsilon_{\mathrm{inj}} + \delta)$.
2. *Minorisation.* $\tilde{\Phi}(\{E \le R\})$ is bounded and $\mathcal{N}(\cdot, dt\,\tilde{Q})$ has a positive continuous density on $\mathcal{V}_h^0$.
3. *Harris* (Hairer–Mattingly 2011, Thm 1.2) : unique π_∞^{dt}, geometric convergence in the $(1+E)$-weighted norm, $\langle E\rangle \le \frac{dt(\varepsilon_{\mathrm{inj}}+\delta)}{1-e^{-2\mu_d dt}}$; the density of $\pi_\infty^{dt} = \pi_\infty^{dt}P$ is a Gaussian convolution. □

The limiter is an analysis device : on $\{E(\mathrm{u}_k) \le E_{\max}\ \forall k\}$ trajectories of Φ and $\tilde{\Phi}$ coincide and $D\tilde{\Phi} = D\Phi$ (TLM unchanged). To do : record the running maximum of E inside the stepper, not only at snapshots.

Backup : TLM proofs

Chain rule. u_l does not depend on ω_k for $l \le k$; $\partial \mathrm{u}_{k+1}/\partial\omega_k = I$; for $l > k+1$, $\partial \mathrm{u}_l/\partial\omega_k = D\Phi(\mathrm{u}_{l-1})\,\partial \mathrm{u}_{l-1}/\partial\omega_k$; $\mathscr{R}_x(0)$ does not depend on $\boldsymbol{\omega}$. □

TLM lemma. By induction $\delta^b_l = \sum_{k<l} D\Phi(x^\circ_{l-1}) \cdots D\Phi(x^\circ_{k+1})\,\omega^b_k$; at $l = n_\mathrm{s}$ this is $\sum_k \mathsf{A}_k \omega^b_k$. For a Gaussian ξ, $\mathrm{Var}(\xi^2) = 2(\mathbb{E}[\xi^2])^2$: relative error $\sqrt{2/n_{\mathrm{probe}}}$. B0 is orbit-constant : L, P, f(dtL), curl^+_h commute with dyadic shifts and Q_ψ is invariant, so Σ^{lin} commutes with the corresponding permutations.

Theorem. $\mathscr{R}_x(\eta_Q^{1/2}\hat{\boldsymbol{\omega}}) = \sum_{k\ge 1} \eta_Q^{k/2} P_k(\hat{\boldsymbol{\omega}})$ with P_k homogeneous of degree k. (1) $P_1 = \mathsf{J}_x$ is linear ; the $k = n_\mathrm{s} - 1$ term of Σ is $dt\ \mathrm{curl}^+_h\, \bar{Q}\, \mathrm{curl}^{+\top}_h$ and curl^+_h maps $\mathcal{V}^0_h = \mathrm{ran}\,\bar{Q}$ onto $\mathcal{R}$, so $\Sigma \succ 0$ on $\mathcal{R}$. (2)–(4) Mean, covariance and third cumulant are sums of expectations of products of P_k's ; a term of total degree odd in $\hat{\boldsymbol{\omega}}$ vanishes since $\hat{\boldsymbol{\omega}} \overset{d}{=} -\hat{\boldsymbol{\omega}}$. Hence they are polynomials in η_Q ; for the third cumulant the lowest candidate order $\eta_Q^{3/2}$ is odd, so it is $O(\eta_Q^2)$. □ **Corollary.** $\gamma_r(\eta_Q^{1/2}\hat{\boldsymbol{\omega}}) = \eta_Q(G_0 + \eta_Q^{1/2} G_1(\hat{\boldsymbol{\omega}}) + \eta_Q G_2(\hat{\boldsymbol{\omega}}) + \dots)$; odd terms have zero mean. □

Mean shift, computable. With $\mathrm{u}_k = x^\circ_k + \eta_Q^{1/2}\delta^{(1)}_k + \eta_Q \delta^{(2)}_k + \dots$: $\delta^{(2)}_{k+1} = D\Phi(x^\circ_k)\delta^{(2)}_k + \frac{1}{2} D^2\Phi(x^\circ_k)[\delta^{(1)}_k, \delta^{(1)}_k]$, hence

$$\mathbb{E}[r \mid x] = \eta_Q\ \mathrm{curl}^+_h \sum_k \mathsf{A}_k\, \tfrac{1}{2}\, \mathbb{E}[D^2\Phi(x^\circ_k)[\delta^{(1)}_k, \delta^{(1)}_k]] + O(\eta_Q^2) \quad \text{(nested JVPs, same probes).}$$

Taylor order is not Wiener-chaos order. By Stroock's formula the first-chaos kernel is $\mathbb{E}[\mathscr{D}_k \mathscr{R}_x(\boldsymbol{\omega})]$, which collects every odd Taylor order : $\mathrm{Cov}(I_1) = \Sigma(x)(1 + O(\eta_Q))$. Chaos components are uncorrelated but not independent, so "Gaussian + non-Gaussian" is not a decomposition of the law ; the rigorous split is the score decomposition relative to p_{ref}. Also : the degree 16^{n_s} of $\mathscr{R}_x$ is a regularity statement only ; all quantitative control comes from the η_Q expansion.

Backup : diffusion and sampler proofs

OU closed form. Variation of constants : $R_\tau = az + \sqrt{2\lambda_j}\sigma \int_0^\tau e^{-\lambda_j(\tau-\tau'')} d\mathcal{W}_{\tau''}$; Itô isometry gives variance $\sigma^2(1-a^2)$.

Tweedie. Differentiate under the integral : $\nabla_R \mathcal{N}(R; az, s) = -\frac{R-az}{s}\mathcal{N}(R; az, s)$; divide by $p_\tau(R \mid x)$ and recognise Bayes' formula for $p(z \mid R, x)$. Finally $R_\tau - az = \sqrt{s}\,\varepsilon$ pathwise. □

Regression. $\mathbb{E}\big[\|f - \mathrm{Y}\|^2\big] = \mathbb{E}\big[\|f - \mathbb{E}[\mathrm{Y} \mid \cdot]\|^2\big] + \mathbb{E}\big[\|\mathbb{E}[\mathrm{Y} \mid \cdot] - \mathrm{Y}\|^2\big]$: the cross term vanishes by the tower property, since $f - \mathbb{E}[\mathrm{Y} \mid \cdot]$ is a function of the conditioning variables. □

Decomposition. Conditioning the rotation lemma : $\mathbb{E}[\varepsilon \mid R_\tau, x] = (\frac{\sqrt{s}}{\sqrt{S_g}})R_\tau^{\mathrm{w}} + (\frac{a\sqrt{\Sigma}}{\sqrt{S_g}})F^\star$. Tweedie :

$s = -(\frac{\sqrt{s}}{\sqrt{S_g}})R_\tau/(\sqrt{s}\sqrt{S_g}) - (\frac{a\sqrt{\Sigma}}{\sqrt{S_g}})F^\star/\sqrt{s}$, and $(\frac{\sqrt{s}}{\sqrt{S_g}})/(\sqrt{s}\sqrt{S_g}) = 1/S_g$. Similarly for $\mathbb{E}[z \mid R_\tau, x]$. □ **Corollary 1.** Under p_{ref}, per coefficient $(z^{\mathrm{w}}, \varepsilon) \sim \mathcal{N}(0, I_2)$, independent across coefficients ; an orthogonal change of basis maps it to $(R_\tau^{\mathrm{w}}, F^{\mathrm{tgt}}) \sim \mathcal{N}(0, I_2)$, so $F^{\mathrm{tgt}} \perp R_\tau^{\mathrm{w}}$ and $F^\star = 0$. □

Bridge. Given z, $R_\tau = a_{\tau|\tau'} R_{\tau'} + \sqrt{s_{\tau|\tau'}}\,\varepsilon''$ (Markov property), $s_{\tau|\tau'} = s_\tau - a^2_{\tau|\tau'} s_{\tau'}$. $(R_{\tau'}, R_\tau)$ is jointly Gaussian with $\mathrm{Cov} = a_{\tau|\tau'} s_{\tau'}$; Gaussian regression gives $\mathbb{E}[R_{\tau'} \mid R_\tau, z] = b_1 R_\tau + b_2 z$, $\mathrm{Var} = s_{\tau'} - a^2_{\tau|\tau'} s^2_{\tau'}/s_\tau = v_{\mathrm{br}}$. At $\tau' = 0$: $b_1 = 0$, $b_2 = 1$, $v_{\mathrm{br}} = 0$. □

Reverse transition and total variance. Given z, the forward process is Markov and independent of x (its noise is independent of (x, ω)), so $p(R_{\tau'} \mid R_\tau, z, x) = p(R_{\tau'} \mid R_\tau, z)$. Laws of total expectation and variance : $\mathbb{E}[R_{\tau'} \mid R_\tau, x] = b_1 R_\tau + b_2 \mathbb{E}[z \mid R_\tau, x]$, $\mathrm{Var}(R_{\tau'} \mid R_\tau, x) = v_{\mathrm{br}} + b_2^2\,\mathrm{Var}(z \mid R_\tau, x)$.

Exactness. Under p_{ref} with $\hat{F}_\theta \equiv 0$, prior, posterior and bridge are Gaussian with the moments used, so every step is exact. Otherwise the per-step covariance error is $b_2^2(\mathrm{Var}(z \mid R_\tau, x) - \bar{\Sigma}s/S_g)$; $b_2 = O(\tau - \tau')$ on interior steps ; at the final jump the posterior variance under p_{ref} is $\bar{\Sigma}/(1+\mathrm{SNR}) \le \bar{\Sigma}/(1+\mathrm{SNR}_{\mathrm{hi}})$.

Backup : diagnostics and hypothesis tests

$F^\star$ **for a Gaussian with errors.** $\mathrm{Cov}(R_\tau) = S_g + a\Xi a$, $\mathbb{E}\big[\varepsilon \mid R_\tau, x\big] = \sqrt{s}(S_g + a\Xi a)^{-1}(R_\tau - a\mathfrak{m})$. Rotation lemma : $F^\star = (\frac{a\sqrt{\Sigma}}{\sqrt{S_g}})^{-1}(\mathbb{E}\big[\varepsilon \mid R_\tau, x\big] - (\frac{\sqrt{s}}{\sqrt{S_g}})R_\tau^{\mathrm{w}})$ with $(\frac{\sqrt{s}}{\sqrt{S_g}})R_\tau^{\mathrm{w}} = \sqrt{s}\, S_g^{-1} R_\tau$. Expand $(S_g + a\Xi a)^{-1} = S_g^{-1} - S_g^{-1} a\Xi a S_g^{-1} + O(\Xi^2)$ and use $(\frac{a\sqrt{\Sigma}}{\sqrt{S_g}})^{-1}\sqrt{s}\, S_g^{-1} a = (\frac{\sqrt{s}}{\sqrt{S_g}})\bar{\Sigma}^{-1/2}$, $a S_g^{-1} R_\tau = (\frac{a\sqrt{\Sigma}}{\sqrt{S_g}})\bar{\Sigma}^{-1/2} R_\tau^{\mathrm{w}}$. □ The weights satisfy $(\frac{\sqrt{s}}{\sqrt{S_g}})(\frac{a\sqrt{\Sigma}}{\sqrt{S_g}}) \le \frac{1}{2}, (\frac{\sqrt{s}}{\sqrt{S_g}}) \le 1$: errors are damped where the SNR is very high or very low.

Empirical linear floor. For each τ of a quadrature grid, $K(\tau) = \arg\min \mathbb{E}\big[\| F^{\mathrm{tgt}} - K R_\tau^{\mathrm{w}} \|^2\big]$ over $\mathcal{G}$-invariant K. With $u = z/\sqrt{\Sigma}$ and ε independent of (u, x) : $\mathbb{E}\big[R^{\mathrm{w}} R^{\mathrm{w}\top}\big] = \mathbb{E}\big[(\frac{a\sqrt{\Sigma}}{\sqrt{S_g}})uu^\top(\frac{a\sqrt{\Sigma}}{\sqrt{S_g}})\big] + \mathbb{E}\big[(\frac{\sqrt{s}}{\sqrt{S_g}})^2\big]$, $\mathbb{E}\big[F^{\mathrm{tgt}} R^{\mathrm{w}\top}\big] = \mathbb{E}\big[(\frac{a\sqrt{\Sigma}}{\sqrt{S_g}})(\frac{\sqrt{s}}{\sqrt{S_g}})\big] - \mathbb{E}\big[(\frac{\sqrt{s}}{\sqrt{S_g}})uu^\top(\frac{a\sqrt{\Sigma}}{\sqrt{S_g}})\big]$: the ε-terms are exact, only u-moments are estimated (training pairs) ; scored on validation pairs with the same $(\tau, \varepsilon, \mathrm{c})$ draws as F_θ. $\mathcal{G} \cong \mathbb{Z}_m^2$ acts freely, so after a 2D DFT over the group index every $\mathcal{G}$-invariant matrix is block diagonal : $|\mathcal{G}|$ Hermitian blocks of size n_{orb} (256 blocks 64×64 at $J = 7$). The same algebra gives Σ^{lin} and $\bar{\Sigma}^z$. Test : excess $\mathcal{J}_{\mathrm{lin}} - \mathcal{J}(\theta)$, paired, s.e. clustered by trajectory ; $z > 3$ rejects H_0 only. Finite sample : the floor's gain is lowered by $\approx 0.7\, n_{\mathrm{orb}}/N$ of itself ; $N/n_{\mathrm{orb}} \ge 5$ required ($\approx$ 100 at $J = 7$).

Size of the non-Gaussian content. For $\Delta \ll \min(\mu_d^{-1}, (\nu k_f^2)^{-1})$: $\chi_3 := k_f \sqrt{\varepsilon_{\mathrm{inj}}/2}\, \Delta^{3/2}$ (skewness), $\chi_{\mathrm{m}} := (k_f u_{\mathrm{rms}} \Delta)\chi_3$ (x-dependent mean shift/sd). Measured from N_{ens} solver members at fixed x : $\chi^2_{\mathrm{NG}} := \frac{1}{d}\sum_i \hat{\mathfrak{m}}_i^2/\hat{\sigma}_i^2 - \frac{1}{N_{\mathrm{ens}}}$. Under $\mathfrak{m} = 0$, $T = N_{\mathrm{ens}} d\, \chi^2_{\mathrm{NG}}$ has $\mathrm{Var}\, T = 2d^2/d_{\mathrm{eff}}$, so a 3σ detection needs $\chi_{\mathrm{NG}} > (18/d_{\mathrm{eff}})^{1/4}/\sqrt{N_{\mathrm{ens}}}$ ($d_{\mathrm{eff}} = n_{\mathrm{orb}}$, conservative). The x-independent part of the mean shift is a lattice term ($2^{-j_{\min}}$-periodic, from the wavelet-diagonal Q_ψ), measured separately and not counted as learnable.

Backup : key references

Wavelets, discretisation

- Cohen, Daubechies, Feauveau, Biorthogonal bases of compactly supported wavelets, CPAM 45, 1992.
- Latto, Resnikoff, Tenenbaum, Connection coefficients of compactly supported wavelets, 1991.
- Beylkin, Operators in bases of compactly supported wavelets, SINUM 29, 1992.
- Dahmen, Wavelet and multiscale methods for operator equations, Acta Numer. 6, 1997.
- Schneider, Vasilyev, Wavelet methods in CFD, Annu. Rev. Fluid Mech. 42, 2010.
- Temam, Navier–Stokes Equations, AMS Chelsea, 2001.
- Lawson, Generalized Runge–Kutta processes, SINUM 4, 1967.
- Hochbruck, Ostermann, Exponential integrators, Acta Numer. 19, 2010.
- Lord, Powell, Shardlow, Computational Stochastic PDEs, CUP, 2014.

Stochastic analysis, physics

- Da Prato, Zabczyk, Stochastic Equations in Infinite Dimensions, CUP, 1992.
- Flandoli, Invariant measures for stochastic NS, NoDEA 1, 1994.
- Hairer, Mattingly, Ergodicity of 2D NS with degenerate forcing, Ann. Math. 164, 2006.
- Hairer, Mattingly, Yet another look at Harris' theorem, 2011.
- Hutzenthaler, Jentzen, Kloeden, Divergence of Euler's method, Proc. R. Soc. A 467, 2011.
- Kuksin, Shirikyan, Mathematics of 2D Turbulence, CUP, 2012.
- Nualart, The Malliavin Calculus and Related Topics, Springer, 2006.
- Boffetta, Ecke, Two-dimensional turbulence, Annu. Rev. Fluid Mech. 44, 2012.

Score-based modelling

- Vincent, Score matching and denoising autoencoders, Neural Comput. 23, 2011.
- Anderson, Reverse-time diffusion equation models, SPA 12, 1982.
- Efron, Tweedie's formula and selection bias, JASA 106, 2011.
- Song et al., Score-based generative modeling through SDEs, ICLR 2021.
- Salimans, Ho, Progressive distillation (v-parameterisation), ICLR 2022.
- Karras et al., Elucidating the design space of diffusion models, NeurIPS 2022.
- Hang et al., Min-SNR weighting, ICCV 2023.
- Guth, Coste, De Bortoli, Mallat, Wavelet score-based generative modeling, NeurIPS 2022.

Learned emulators

- Kochkov et al., ML-accelerated CFD, PNAS 118, 2021.
- Li et al., Fourier neural operator, ICLR 2021.
- Stachenfeld et al., Learned coarse models for turbulence, ICLR 2022.
- Lippe et al., PDE-Refiner, NeurIPS 2023.
- Kohl, Um, Thuerey, Autoregressive conditional diffusion for turbulence, arXiv :2309.01745, 2023.
- Price et al., Probabilistic weather forecasting with ML (GenCast), Nature 637, 2025.